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A parametric head geometry model accounting for variation among adolescent and young adult populations

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Highlights

- Head CT scans of 101 subjects between 14 and 25 years of age were used to develop a statistical head geometry model.
- Skull and scalp statistical geometry models accounts for size and shape variations among the adolescent and young adult population were developed as functions of age, sex, stature, BMI, head circumference, and trignon-to-top distance.
- The statistical geometry models account for a high percentage of morphological variations in scalp, outer skull, inner skull, and skull thickness, and may serve as the geometric basis to develop individualized head finite element models for injury assessment and design of head-borne equipment.

Abstract

Background and objective

Modeling the size and shape of human skull and scalp is essential for head injury assessment, design of helmets and head-borne equipment, and many other safety applications. Finite element (FE) head models are important tools to assess injury risks and design personal protective equipment. However, current FE head models are mainly developed based on the midsize male, failing to account for the significant morphological variation that exists in the skull and brain. The objective of this study was to develop a statistical head geometry model that accounts for size and shape variations among the adolescent and young adult population.

Methods

To represent subject-specific geometry using a homologous mesh, threshold-based segmentation of head CT scans of 101 subjects between 14 and 25 years of age was performed, followed by landmarking, mesh morphing, and projection. Skull and scalp statistical geometry models were then developed as functions of age, sex, stature, BMI, head length, head breadth, and trignon-to-top of head using generalized Procrustes analysis (GPA), principal component analysis (PCA) and multivariate regression analysis.

Results

The statistical geometry models account for a high percentage of morphological variations in scalp geometry ($R^2=0.63$), outer skull geometry ($R^2=0.66$), inner skull geometry ($R^2=0.55$), and skull thickness (error < 1 mm)

Conclusions

Skull and scalp statistical geometry models accounts for size and shape variations among the adolescent and young adult population were developed as functions of subject covariates. These models may serve as the geometric basis to develop individualized head FE models for injury assessment and design of head-borne equipment.



Keywords

Head geometry; Skull geometry; Statistical model; Human variation; Traumatic brain injury

1. Introduction

1.1. Statistical shape model of human head

Statistical Shape Models (SSMs) have been employed to characterize human geometric variability and identify variations within a large sample of 3D geometric bones and organs. Studies include SSMs of the rib cage [1], [2], [3], femur [4], [5], [6], tibia [7,8], liver [9,10], spleen [11], and the external body shape [12], [13], [14]. These models have served as the geometrical basis for building subject-specific or parametric human finite element (FE) models for various applications, such as injury assessment, computer-aided surgery, personal protective equipment design, and many other designs that interact with human. In terms of the head/skull geometry model, Urban et al. [15] assessed adult outer skull morphology changes associated with age and sex using 122 subject head CT scans from samples between 20 and 100 years of age. Danckaers et al. [16] developed a statistical shape model combining the inner skull, outer skull, and scalp geometries using MRI scans from 85 subjects. However, this model did not correlate morphologic variations with standard anthropometric dimensions. Some other studies quantified variations in the scalp and face [17], [18], [19], but did not include the skull and skull thickness, making detailed biomechanical analysis involving skull and brain difficult.

1.2. Head injury assessment

In 2014, traumatic brain injuries (TBIs) accounted for approximately 2.87 million emergency department visits, hospitalizations, and deaths in the United States, and of these incidents, individuals aged 15-24 years old experienced the highest rate of TBI-related hospitalizations resulting from motor vehicle crashes [20]. Sports-related activities are also a frequent cause of TBIs in adolescents; individuals aged 5-18 years old accounted for 65% of emergency room visits of all sports/recreational-related TBIs in 2011 [21]. The rate of concussion was reported to be higher and recovery time longer for high school vs. athletes aged 20-22, and neurocognitive deficits in adolescent athletes may also persist well after the self-reported resolution of concussion symptoms [22]. The immaturity of the developing central nervous system, a larger head-to-body ratio, thinner cranial bones, a larger subarachnoid space, and variations in cerebral blood volume have all been cited as potential risk factors specific to adolescents aged 14-18 years, accounting for increased vulnerability to brain injuries in adolescents when compared to young adults 20-22 years old [22].

Head FE models are important tools for evaluating skull, brain, and other tissue-level responses [23]. However, such models typically focused on a single subject close to the 50th percentile male and failed to account for morphological variations in the population. This poses a risk for the generalizability of these injury assessment studies given the lack of representation for vulnerable populations that are at higher risk of head injuries. A head SSM including the detailed anatomy of head skull and brain can be employed to address this gap by enabling the rapid generation of

individualized FE models for adolescents and young adults based on common anthropometric measurements.

1.3. Design of helmets, head-borne equipment and other safety applications

In recent years, head SSMs have also been employed in the industry for the design of headgear. Compared to traditional anthropometric body measurements used for headgear design, head shape models provide individualized 3D geometry, allowing for greater design efficiency, fit, and potentially functionality. Previous studies have demonstrated that the use of SSM provides valuable benefits to the assessment of helmet fit [24], as well as the design of helmets [25], respirators [26] and Brain-Computer Interface (BCI) headsets [27]. Additionally, head FE models are often employed in impact simulations to assess the effectiveness of commercial head protective equipment [28], [29], [30], [31].

1.4. Research objective

The objective of this study was to develop a SSM of the head (including the inner skull, outer skull, and scalp) accounting for morphological variations by age, sex, stature, BMI, and standard head anthropometric measures for adolescents and young adults. Head CT scans of 101 subjects between 14 and 25 years of age were analyzed. The statistical geometry models account for a high percentage of morphological variations in scalp, outer skull, inner skull, and skull thickness. This study was motivated by the lack of consideration of morphological variation in head injury assessment tools; a failure to provide safety equity for the population as a whole. This SSM model will also serve as the geometric basis to develop individualized head FE models for the design of helmets, head-borne equipment and other safety applications.

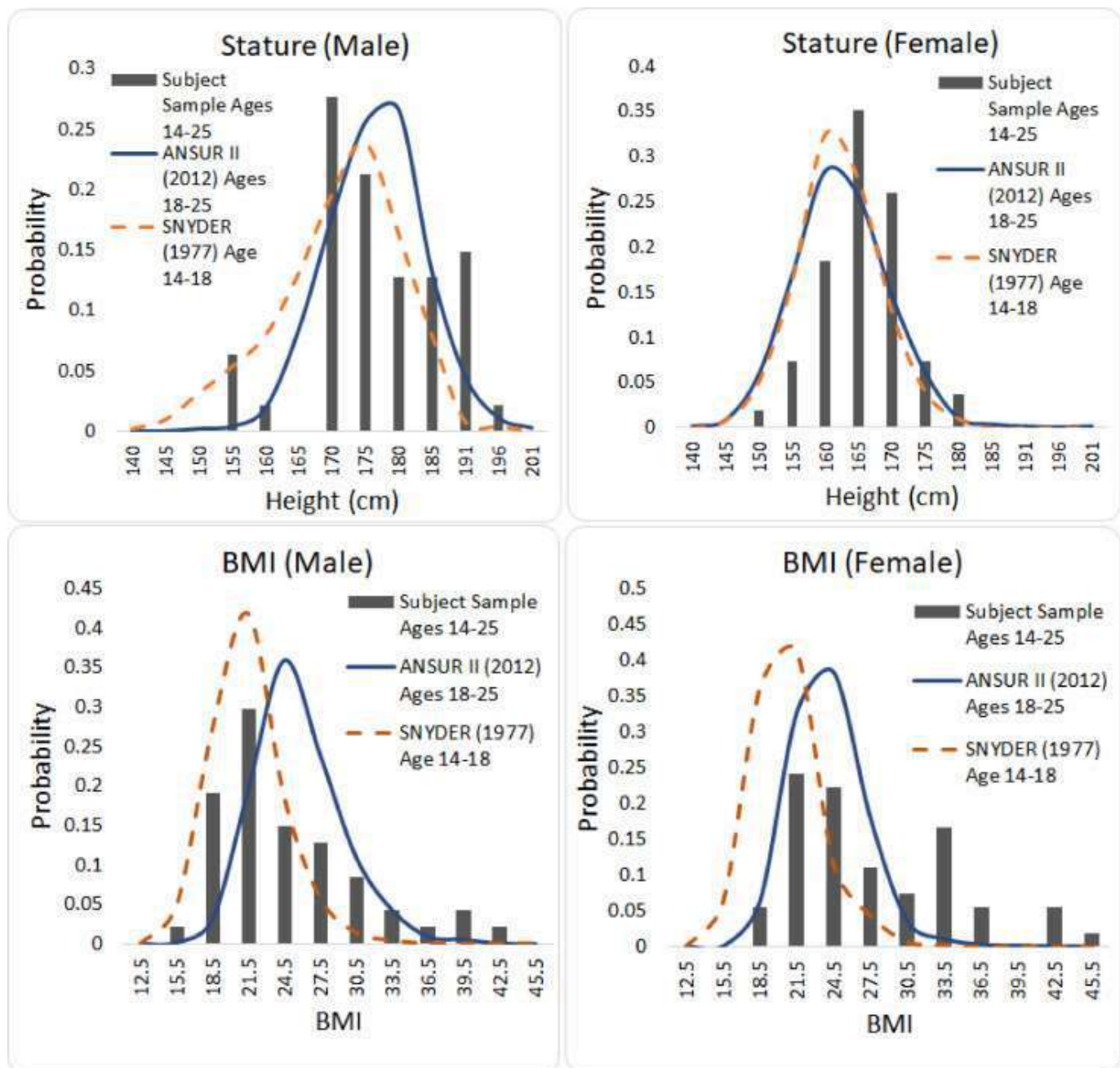
2. Methods

2.1. Data source

Head CT scans from 101 subjects (47 male and 54 female) were obtained from the University of Michigan Health System. The study protocol was approved by the University of Michigan Institutional Review Board (IRB) with an approved case ID of HUM00004842. The age range was 14 to 25 years with a mean of 19.7 years for male and 19.9 years for female. Participants' body mass index (BMI) values ranged from 15.6 to 46.8 kg/m² with a mean of 24.9 kg/m² for male and 27.8 kg/m² for female. The stature range was 152.5–195.6 cm for male and 152.4–182.9 cm for female. Additional head measurements, including head breadth, length, circumference, and trignon-to-top of head, were computed directly from subject scalp CT segmentation based on definitions from Anthropometric Survey of U.S. Army Personnel (ANSUR-II) [32].

Fig. 1 illustrates the height and BMI distributions of the study sample, relative to samples from the U.S. Army personnel, and a U.S. child anthropometry survey. Data for individuals from 2010–2012

ANSUR-II [32] that are within the age range of our study sample (18-25) were compared. Similarly, data for individuals that are within the age range of our current study (14-18) were selected from the anthropometric survey data gathered by Snyder et al. [33].



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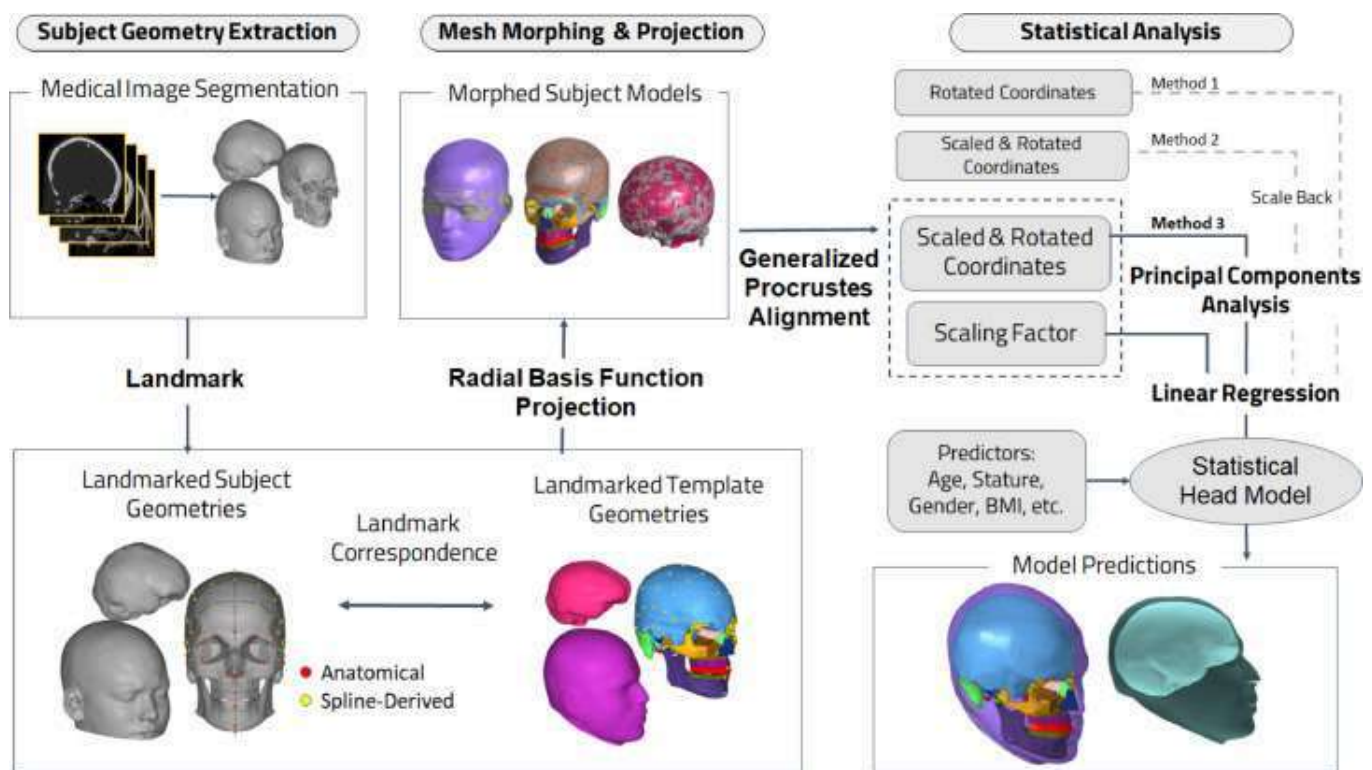
Fig. 1. Distributions of stature and body mass index for males and females in age-matched samples from ANSUR II (2021) and Snyder (1977) datasets, and the current sample.

The distributions of stature and BMI by sex and age groups defined at 14-19 and 20-25 years of age, were contrasted with corresponding percentiles from 2011 to 2014 NHANES report [34]. For male, the 5th, 50th, and 95th percentile stature values (158, 173, 188 cm respectively for ages 14-19; 169, 180, and 193 cm for ages 20-25) matched the U.S. population for the same age cohorts. For female

with age of 14-19, our study cohort had slightly higher stature relative to NHANES for the 5th and 50th percentile values (156 and 166 cm), while the 95th percentile value (172 cm) matched the NHANES data. Female subjects with age 20-25 in our study cohort were slightly taller compared to the general U.S. population (155, 166, 178 cm corresponding to 5th, 50th, and 95th percentile values).

2.2. Method overview

[Fig.2](#) illustrates an overview of the methodology we employed to develop the statistical head geometry model for adolescents and young adults. First, surface geometries of inner skull, outer skull, and scalp of each sampled subject were extracted through threshold-based image segmentation. The segmented data were aligned to an anatomically derived coordinate system and landmarked using a semi-automated procedure. A template mesh was then morphed and projected to each segmented surface mesh. The morphed/projected subject-specific meshes were analyzed using three statistical methods. The first method used generalized Procrustes analysis (GPA) without scaling, which aligned subject-specific models without accounting for scale. The aligned and unscaled nodal coordinates were then analyzed with principal component analysis (PCA) and regression to generate the final statistical model. The second method involved GPA with scaling, which aligned subject-specific models while accounting for a scaling factor. The scaling factor for each subject was recorded and used to recover their original scale immediately following the GPA step. PCA and regression were then performed on the aligned and re-scaled nodal coordinates. This is the approach used to generate statistical models in many previous studies [\[1,5\]](#). The third and chosen method for this study, also involved GPA with scaling, accounting for a scaling component during alignment. However, this method separated the shape and size effects by generating separate regression models for predicting the scaled nodal coordinates and the scaling factor separately. The regression models for size and shape were then combined in the final model. Because the third method is the chosen method in this study, we only described the details of the third method in [section2.5](#) while the results of the other methods were used for comparison purpose.



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Fig. 2. Methodology for generating a statistical shape model of the inner and outer skull and scalp.

2.3. Medical image segmentation and landmark protocol

Image segmentation was performed on subject CT images (512×512 resolution) using Mimics (Materialise, Belgium). Skull and scalp geometries were extracted using the Mimics default thresholds for bone and scalp geometries, which are 226 and -400, respectively. The segmented skull and scalp surfaces were then aligned to a coordinate system according to the Frankfurt plane, which is defined by the horizontal plane passing through the left and right trignon, and right infraorbitale landmarks.

A total of 102 anatomical and spline-derived landmarks were identified on the outer skull of each subject using HyperMesh 14.0 (Altair, U.S.). Thirty anatomically-derived head landmarks [35,36] were manually digitized along the facial and posterior regions of the outer skull. Another 72 landmarks were digitized along the outer skull surface using a cubic-spline method. Details of the semi-automatic landmark protocol, as well as definitions of the anatomically derived landmarks are documented in [Appendix A](#).

2.4. Mesh morphing and projection

A set of template surface meshes of the inner skull, outer skull, and scalp from the midsize male model from the Global Human Body Models Consortium (GHBMC v4.4) [37] were mapped to the

segmented skull and scalp surfaces for each sampled subject. To do so, the same set of landmarks identified for each subject was also identified on the template meshes. The template meshes are then morphed based on landmarks of subject-specific geometries using a radial basis function with thin plate spline (RBF-TPS). The RBF-TPS technique ensures specific template vertices are located at corresponding landmark locations across all subject-specific meshes. This technique has been described in detail in our previous studies [38,39]. Following the initial morphing, the morphed template meshes of scalp, inner and outer skull layers were then projected onto subject geometries to ensure a more accurate match to the geometric details. Consequently, subject-specific meshes, consisting of the inner skull, outer skull, and scalp surfaces, were generated for each subject. Landmarks were not defined on the inner skull and scalp, because the landmarks on the outer skull alone with mesh morphing and projection lead to consistent mesh quality and geometry accuracy in all three layers (i.e., inner skull, outer skull, and scalp).

2.5. Statistical analysis

The subject-specific mesh data were statistically analyzed using three different approaches (described in [Section 2.2](#)) involving GPA, PCA and regression analysis.

2.5.1. Generalized Procrustes analysis

Let $\mathbf{P} = \{\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_{n_1}\}$, $\mathbf{p}_i \in \mathbb{R}^{m \times 3}$ denote the nodal coordinates of n_1 subjects. For the i -th subject $\mathbf{p}_i = [\mathbf{p}_{i,1}, \mathbf{p}_{i,2}, \dots, \mathbf{p}_{i,m}]^T$ with m nodes, the j -th node $\mathbf{p}_{i,j} = [\mathbf{x}_{i,j}, \mathbf{y}_{i,j}, \mathbf{z}_{i,j}] \in \mathbb{R}^3$ contains $\mathbf{x}, \mathbf{y}, \mathbf{z}$ coordinates. Throughout the paper, we denote scalars by lowercase letters, vectors by lowercase boldface letters, and matrices, sets of vectors by uppercase boldface letters. GPA was performed to align all the subjects by scaling, translating and rotating the nodes. The i -th subject was aligned with the template mesh $\mathbf{p}_{\text{base}} \in \mathbb{R}^{m \times 3}$ as the following steps:

- 1 Compute the centroid of the baseline model $\bar{\mathbf{p}}_{\text{base}} = \frac{1}{m} \sum_{j=1}^m \mathbf{p}_{\text{base},j}$ and the centroid of the i -th subject $\bar{\mathbf{p}}_i = \frac{1}{m} \sum_{j=1}^m \mathbf{p}_{i,j}$
- 2 Re-scale the i th subject to the same size of baseline model, the scaling factor $\mathbf{s}_i$ is calculated as

$$\mathbf{s}_i = \sqrt{\frac{\text{tr}((\mathbf{p}_{\text{base}} - \bar{\mathbf{p}}_{\text{base}})^T (\mathbf{p}_{\text{base}} - \bar{\mathbf{p}}_{\text{base}}))}{\text{tr}((\mathbf{p}_i - \bar{\mathbf{p}}_i)^T (\mathbf{p}_i - \bar{\mathbf{p}}_i))}} \quad (1)$$

where $\text{tr}(\cdot)$ is the trace.

- 3 Find the rotation matrix $\mathbf{R}_i \in \mathbb{R}^{3 \times 3}$ that minimizes the Euclidean norm between the transformed i th subject and the baseline model

$$\mathbf{R}_i = \underset{\mathbf{R}_i}{\text{argmin}} \left\| \mathbf{s}_i \mathbf{p}_i \mathbf{R}_i + \mathbf{I} (\bar{\mathbf{p}}_{\text{base}} - \bar{\mathbf{p}}_i) - \mathbf{p}_{\text{base}} \right\|_2^2 \quad (2)$$

where $\mathbf{I}$ is a $(m \times 1)$ unit vector, the aligned coordinates are formulated as

$$\mathbf{p}_i^A = \mathbf{s}_i \mathbf{p}_i \mathbf{R}_i + \mathbf{I} (\bar{\mathbf{p}}_{\text{base}} - \bar{\mathbf{p}}_i) \quad (3)$$

2.5.2. Principal component analysis and linear regression

After GPA, the three-dimensional nodal coordinate data $\mathbf{P}^A = \{\mathbf{p}_1^A, \mathbf{p}_2^A, \dots, \mathbf{p}_{n_1}^A\}$, $\mathbf{p}_i^A \in \mathbb{R}^{m \times 3}$ were re-organized as a matrix $\mathbf{X} \in \mathbb{R}^{n_1 \times 3m}$

$$\mathbf{X} = \begin{bmatrix} x_{1,1}^A, y_{1,1}^A, z_{1,1}^A & \cdots & x_{1,m}^A, y_{1,m}^A, z_{1,m}^A \\ \vdots & \ddots & \vdots \\ x_{n,1}^A, y_{n,1}^A, z_{n,1}^A & \cdots & x_{n,m}^A, y_{n,m}^A, z_{n,m}^A \end{bmatrix}$$

where $\mathbf{p}_{i,j}^A = [x_{i,j}, y_{i,j}, z_{i,j}] \in \mathbb{R}^3$ for the j -th node in $\mathbf{p}_i^A$.

PCA was used to quantify primary modes of variation in the dataset by extracting principal components (PCs), a set of linearly independent variables representing maximal variation of each layer, while minimizing the dimensionality. Based on PCA, $\mathbf{X}$ can be decomposed as $\mathbf{X} = \mathbf{S}_k \mathbf{P}_k + \epsilon$, where $\mathbf{S}_k \in \mathbb{R}^{n_1 \times k}$ is the PC scores corresponding to the first k PCs, $\mathbf{P}_k \in \mathbb{R}^{k \times 3m}$ is the coefficient matrix, and ϵ is the error.

After PCA, two linear regression models were developed to quantify head shape and size effects respectively. Linear regression models were first used to predict PC scores of the scalp, outer skull and inner skull layers using different combinations of a set of subject covariates as input parameters, which are age, stature, sex, BMI, head breadth, head length, trignon-to-top of head. The training input data is then formulated as $\mathbf{F}_{\text{train}} \in \mathbb{R}^{n_1 \times 8}$ with 7 covariates and intercept. The linear regression model is then constructed as

$$\mathbf{S}_k = \mathbf{F}_{\text{train}} \mathbf{C}_k + \boldsymbol{\xi} \quad (4)$$

where $\mathbf{C}_k \in \mathbb{R}^{8 \times k}$ is the coefficients of the linear regression model for the first k PCs, and $\boldsymbol{\xi}$ is the residual error. An ordinary least squares regression is used for linear regression by minimizing the sum of the squared errors. The coefficient $\mathbf{C}_k$ is estimated as

$$\mathbf{C}_k = (\mathbf{F}_{\text{train}}^T \mathbf{F}_{\text{train}})^{-1} \mathbf{F}_{\text{train}}^T \mathbf{S}_k \quad (5)$$

Following the same procedure, the scaling factor $\mathbf{s} \in \mathbb{R}^{n_1 \times 1}$ was also predicted as a function of the selected input parameters. The linear regression model for the scale is

$$\mathbf{s} = \mathbf{F}_{\text{train}} \mathbf{C}_{ks} + \boldsymbol{\xi}_s \quad (6)$$

with coefficients $\mathbf{C}_{ks} = (\mathbf{F}_{\text{train}}^T \mathbf{F}_{\text{train}})^{-1} \mathbf{F}_{\text{train}}^T \mathbf{s}$ and error $\boldsymbol{\xi}_s$.

2.5.3. Model prediction and evaluation

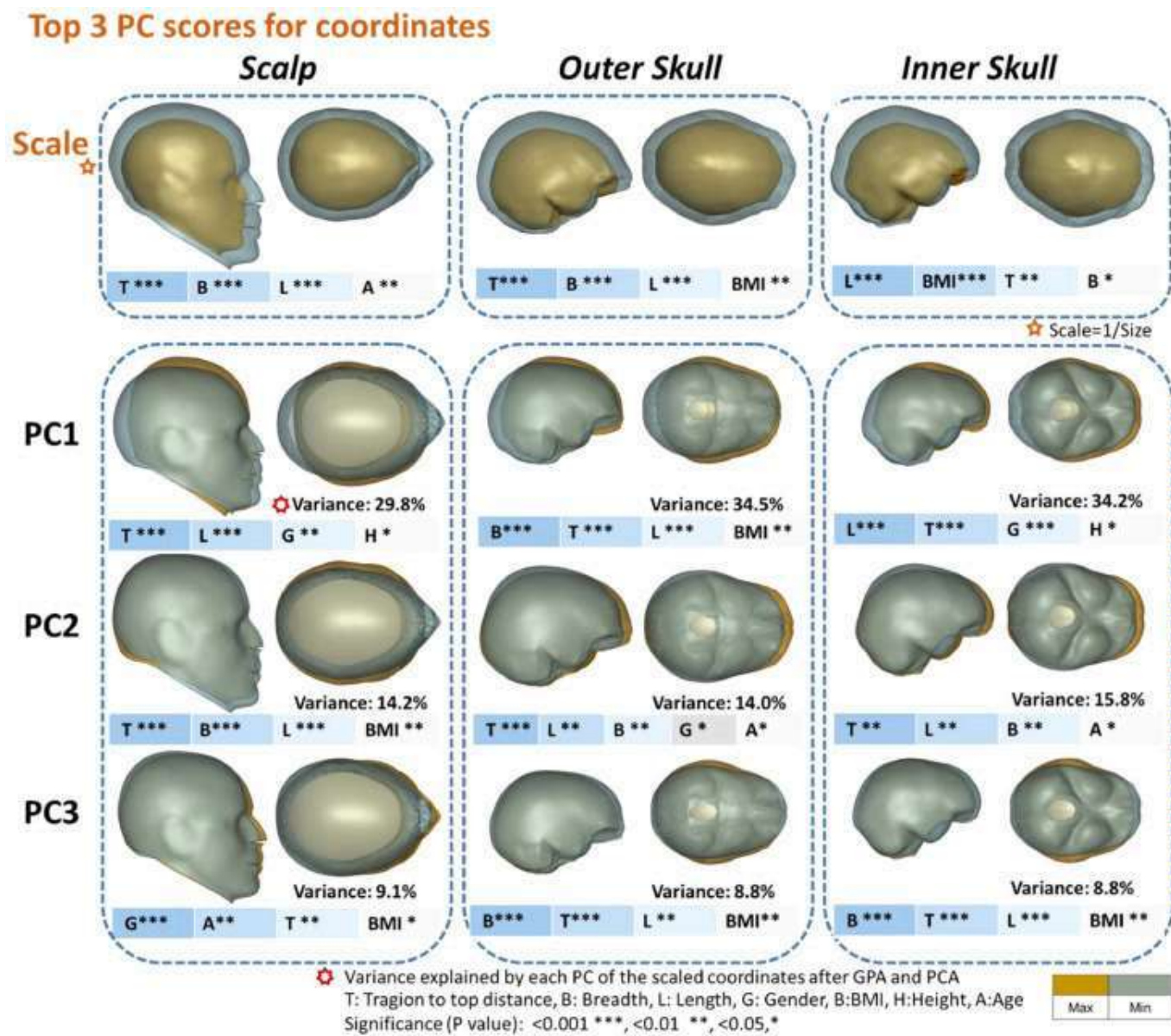
After building the linear regression model for the shape and size, the final geometry model was predicted for n_2 test subjects with test input data $\mathbf{F}_{\text{test}} \in \mathbb{R}^{n_2 \times 8}$. The predicted PC scores $\hat{\mathbf{S}}_{k,\text{test}} = \mathbf{F}_{\text{test}} \mathbf{C}_k$ and the scale $\hat{\mathbf{s}}_{\text{test}} = \mathbf{F}_{\text{test}} \mathbf{C}_{ks}$ were calculated separately using the linear models in Eq.(4) and (5). By reversing PCA, the predicted coordinate matrix $\hat{\mathbf{X}}_{\text{test}} = \hat{\mathbf{S}}_{k,\text{test}} \mathbf{P}_k$ were reconstructed. For k -th test subject $k \in \{1, \dots, n_1\}$, the j -th node is extracted from $\hat{\mathbf{X}}_{\text{test}}$ as $\hat{\mathbf{p}}_{k,j}^A = [\hat{x}_{k,j}^A, \hat{y}_{k,j}^A, \hat{z}_{k,j}^A] \in \mathbb{R}^3$, the prediction of the j -th node is then calculated as $\hat{\mathbf{p}}_{k,j} = \frac{1}{\hat{s}_{\text{test},k}} \hat{\mathbf{p}}_{k,j}^A$.

The coefficient of determination R^2 is used to evaluate the accuracy of the linear regression model by comparing the actual test coordinates $\mathbf{P}_{\text{test}} = \{\mathbf{p}_{\text{test},1}, \mathbf{p}_{\text{test},2}, \dots, \mathbf{p}_{\text{test},n_2}\}$, $\mathbf{p}_k \in \mathbb{R}^{m \times 3}$ and $\hat{\mathbf{P}}_{\text{test}} = \{\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_2, \dots, \hat{\mathbf{p}}_{n_2}\}$, $\hat{\mathbf{p}}_k \in \mathbb{R}^{m \times 3}$.

3. Results

3.1. Size and shape analysis

All 100 PCs were retained from PCA, which accounted for all variation in the participant sample. Visualization of the shape and size variations represented by the first three PCs and scaling factor is shown in Fig.3. The percentages of variance explained by the scaling factor as well as the first three PCs for the scalp, outer skull, and inner skull are also reported in Fig.3. Note that PCA was performed on scaled coordinates following GPA, as mentioned in section 2.5. Therefore, the percentages of variance explained (50.2%, 49.6%, and 50.9% for the scalp, outer skull, and inner skull, respectively) only apply to the shape variation and do not account for the size variation explained by scaling.



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Fig. 3. Visualization and prediction results for the first three principal components and scaling factor effects on the scalp, outer skull, and inner skull geometry.

3.2. Regression results

The set of parameters used in linear regression were selected from subject covariates (age, BMI, sex, stature), and computed head measurements (head circumference, head length, head breadth, and trigion-to-top of head). The R^2 values of statistical head models generated with different parameter combinations are reported in [Table 1](#). The statistical head model generated using all parameters achieved the highest R^2 values, followed by the model generated with all subject parameters except head circumference. Given that head breadth and length are highly correlated with head circumference, head circumference was not used in the final regression model. Based on the R^2

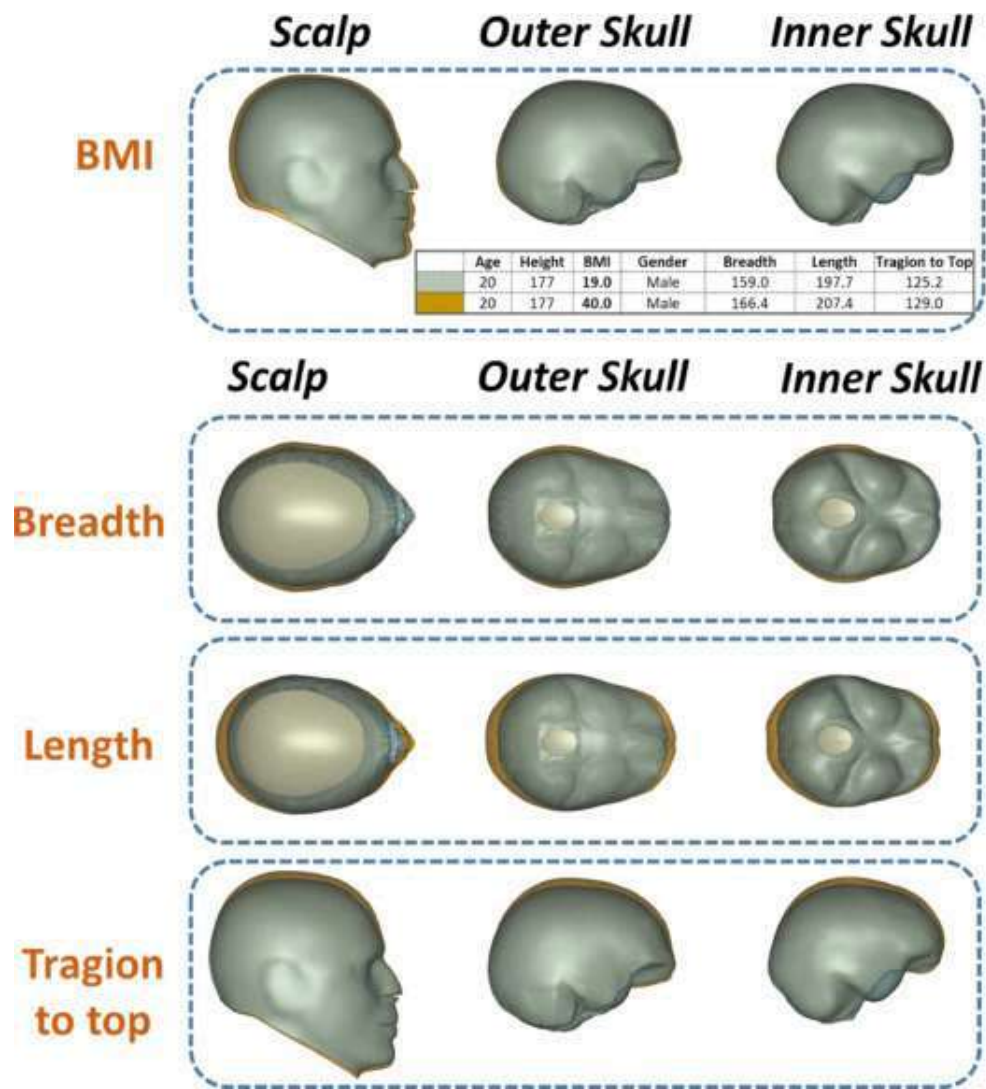
values, the final statistical model accounted for 66% variation of the scalp, 66% variation of the outer skull, and 56% variation of the inner skull by age, BMI, sex, stature, head length, head breadth, and tragon-to-top of head.

Table 1. R² values of the PCAR models with various combinations of input parameters.

	SC	SC+C	SC+B	SC+L	SC+B+L	SC+T	SC+C+T	SC+B+L+T	All
Scalp	0.32	0.54	0.44	0.50	0.59	0.49	0.60	0.66	0.68
Outer skull	0.39	0.57	0.48	0.55	0.60	0.52	0.62	0.66	0.67
Inner skull	0.30	0.47	0.39	0.45	0.50	0.42	0.52	0.56	0.57

SC: subject covariates including age, sex, stature, and BMI, C: head circumference, B: head breath, L: head length, T: tragon to top distance.

Fig.4 shows the geometric variations for male participants with ±1 standard deviation for head breath, head length, and tragon-to-top of head, respectively, while keeping all other input parameters at the sampled population mean.



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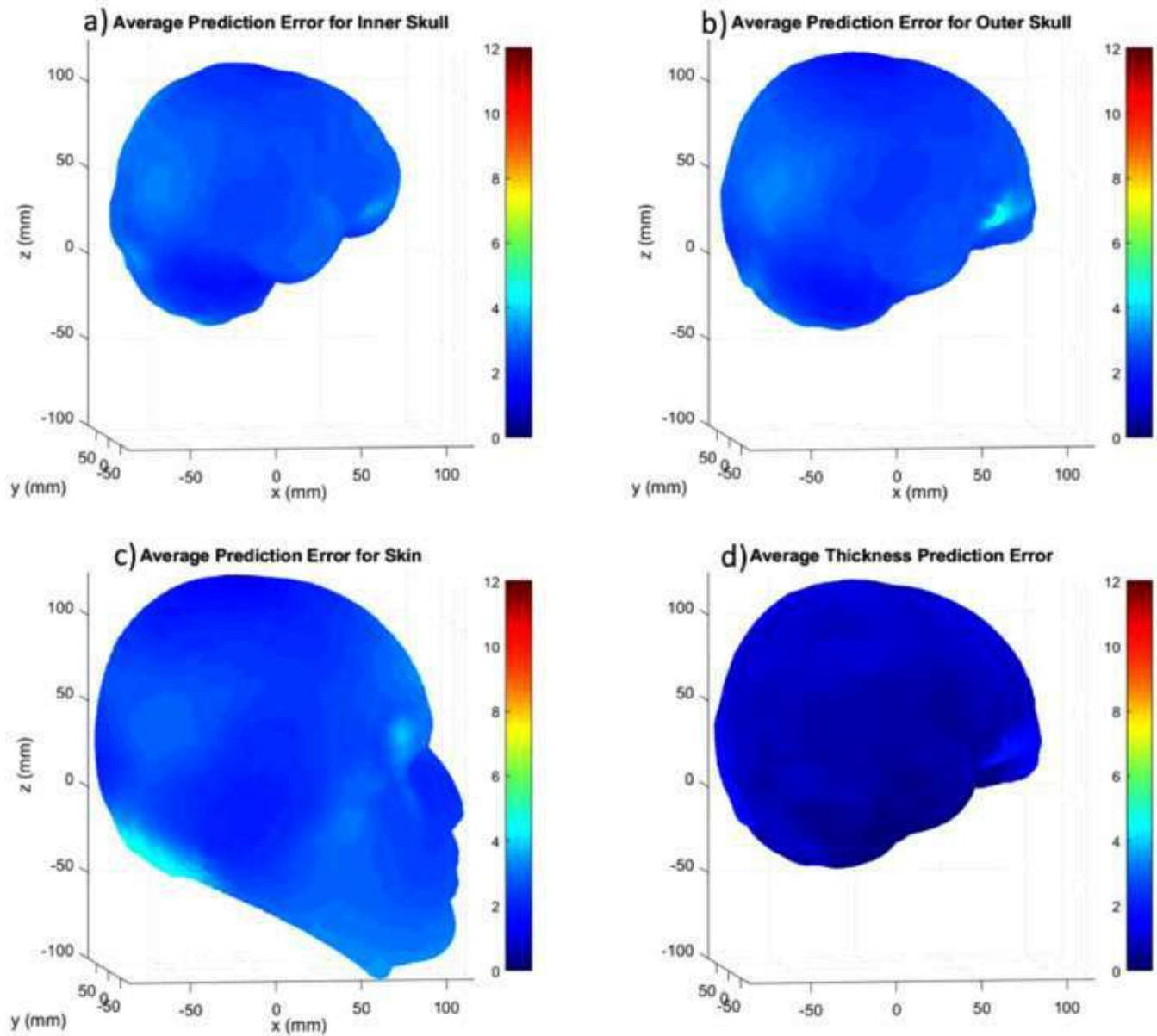
Fig. 4. Outputs from the scalp, outer skull, and inner skull statistical shape models showing a range (± 1 standard deviation from mean) of effects from BMI, head breadth and length, and tragon-to-top head measures on each shape.

The significance level of each parameter on the first three PCs are reported in Fig. 3. Tragon-to-top of head ($p < 0.001$), head breadth ($p < 0.001$), head length ($p < 0.001$), sex ($p < 0.01$), and height ($p < 0.05$) were significant in PC 1. Tragon-to-top of head, head breadth, head length, BMI were also significant in PC 2, PC 3, and scaling factors. Sex was significant in PC 2 and 3. Age was significant in PC 2.

3.3. Skull thickness

Skull thickness was measured by the Euclidean distance between corresponding nodes on the outer and inner skull layers. Thickness prediction error was calculated by computing the difference between the skull thickness measured from the predicted subject geometries and those measured

from the subject-specific geometries. Fig.5 shows the average error associated with each node of inner skull, outer skull, and skull thickness prediction across all subjects. The prediction error for skull thickness was within 1 mm at all node locations in all subjects.



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Fig. 5. Average model prediction error (mm) distributions associated with each node across the sampled subjects for the inner skull, outer skull, and skin layers as well as the skull thickness. The statistical geometry model was used to predict the location of each node in the inner skull, outer skull, and skin, as well as the skull thickness associated with each node on the outer skull. The error between the model prediction and the CT-based subject geometry was recorded for each node on each subject. The average nodal location error distributions for the inner skull, outer skull, and skin are shown in Fig.5a), 5b), and 5c), respectively, and the average skull thickness error is shown in Fig.5d).

4. Discussion

4.1. Separation of shape and scaling predictions

R^2 values for the regression models of the scalp, outer skull, and inner skull using the three statistical methods mentioned in [Section 2.2](#) are reported in [Table 2](#). Overall, Method 3 (scaling and shape separated PCA & regression) explained the most variance across all layers, followed by Method 2 (scaling and shape combined in PCA & regression), both of which explained significantly more variance than Method 1 (no scaling in GPA). The improved performance of Methods 2 and 3 can be attributed to the inclusion of scaling in GPA, allowing for better alignment between subject geometry coordinates. Method 3 differs from Method 2 by further separating the shape and size effects through two regression models. When PCA is applied to originally sized coordinates, as in Methods 1 and 2, the first PC is dominated by the size effect, since it is the most significant contributor to the overall variance. Separating the scaling from the shape effects allows PCA to better quantify shape variation, leading to better prediction results by the statistical model. Historically, head injury assessment tools only focus on the midsize male, while other sizes of head models are often scaled from the midsize male. This study showed that the size scaling could account for a considerable amount of head geometry variations, but the head shape variation is also significant, which should be properly considered when developing head injury assessment tools accounting for a diverse population.

Table 2. R^2 values of the three methods for building the statistical head geometry models.

	Scalp	Outer Skull	Inner Skull
Method 1: No scaling in GPA	0.65	0.56	0.50
Method 2: Scaling and shape combined in PCAR	0.67	0.59	0.53
Method 3: Scaling and shape separated in PCAR (Model R^2/Scaling R^2)	0.66/0.93	0.58/0.83	0.53/0.71

4.2. Head morphology

For 3D model generation purposes, computing the standard head dimensions (tragon-to-top of head, head breadth, and head length) provided not only direct bounds for external (scalp) models but also strong predictions for intracranial structures, specifically the outer and inner skull geometry. Previous studies [\[15,25,40,41\]](#) found correlation between subject covariates, namely age and sex, and head geometry. While our results indicate that covariates (age and BMI) are significantly correlated with specific principal components, the addition of calculated head measurements capture more geometric variation among the sample dataset. Subject covariates (age, sex, stature, and BMI) can only account for 30-39% of the geometry variations in the three layers, while adding head measures (head breath, length, and tragon-to-top of head) can account

for 56-66% of those variations. Specifically, BMI was discovered to significantly affect the outer skull geometry and skull thickness, while age was most significant in the scaling factor of the scalp, although present across all three geometric layers. Nevertheless, our results showed that subject covariates can only account for a limited portion of head geometric variations, which is much less than those for the long bones (e.g. ribs, femur, etc.) reported by our previous studies [2,5].

4.3. Limitations

The manual collection of anatomical landmarks is time-consuming and may lead to varying repeatability in landmark extraction across individuals. To mitigate this error, cubic spline landmarking allows even distribution of landmarks across the skull area, while the mesh morphing/projection step consistently distributes the same template meshes to the skull. It is not clear how landmark repeatability may affect the accuracy and power of the statistical model. However, we expect that such effect should be much smaller than the geometry variations in the population. Another limitation of this study is the lack of facial data in the sampled CT scans. Because most sampled CT scans only focused on the brain, facial bones are often out of the view. As a result, if facial bone landmarks are not available, facial bone geometries were morphed based on the other landmarks identified. The current CT data also provides limited detailed brain geometry, although the inner skull geometry should be highly correlated with brain size and shape. MRI data analysis would be needed if detailed brain geometry model (e.g., sulci, pia, falx, tentorium, ventricles, etc.) is needed. Furthermore, we had a limited age range of the dataset, as we focused on adolescents and young adults in this study. Adding data from middle-aged samples would further explain variation, particularly the effect of age, in fully mature adult skulls. In addition, this study only focused on morphological variations, but not the biomechanical/material variations in human tissues. However, a recent study [42] has shown that morphological variations alone are highly correlated to tissue level dynamic brain responses. Therefore, the statistical model developed in this study would be valuable for future brain injury assessment.

5. Conclusion

In this study, we developed a statistical head geometry model that accounts for size and shape variations among the adolescent and young adult population. To represent subject-specific geometry using a homologous mesh, threshold-based segmentation of head CT scans of 101 subjects between 14 and 25 years of age was performed, followed by landmarking, mesh morphing, and projection. Skull and scalp statistical geometry models were then developed as functions of age, sex, stature, BMI, head length, head breadth, and trignon-to-top of head using GPA, PCA and multivariate regression analysis. The statistical geometry models account for a high percentage of morphological variations in scalp geometry ($R^2=0.63$), outer skull geometry ($R^2=0.66$), inner skull geometry ($R^2=0.55$), and skull thickness (error < 1 mm), and may serve as the geometric basis to develop individualized head FE models for injury assessment, headgear design, and other safety applications.

Declaration of Competing Interest

There is no conflict of interest associated with this study.

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Appendix B. Supplementary materials

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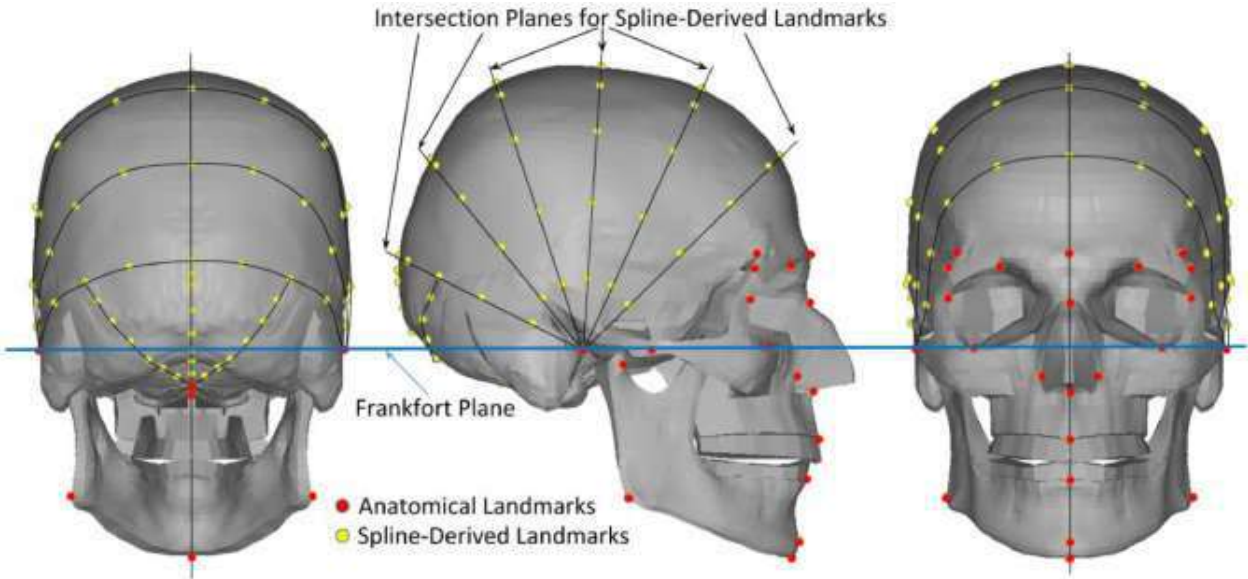
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Appendix A. Landmarking Protocol

A total of 102 landmarks were identified for each subject using Hypermesh (Altair, Troy, MI). Among them, 30 landmarks are at locations that can be identified based on skull anatomy. This set of landmarks were defined as “Anatomical landmarks”. Another set of 72 landmarks on the outer skull were identified semi-automatically using a cubic-spline method. These surface landmarks were defined as “Spline-derived landmarks” and are evenly distributed on the skull outer surface. A landmarked skull is illustrated in [Fig.A1](#), and a list of anatomical landmarks and their associated definitions can be found in [TableA1](#).



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Fig. A1. Illustration of landmark locations.

Table A1. Landmark location definitions.

Name	Definition
Porion (L, R)	The most lateral point on the roof of the bony external acoustic meatus, vertically over the middle of the meatus (ear canal)
Infraorbitale (L, R)	The most inferior point of each infraorbital rim
Glabella	The most anterior point of frontal bone between supraorbital in the mid-sagittal plane
Subnasale	The lowest point of lower anterior nasal aperture in the mid-sagittal plane
Sellion	The point of maximal curvature of the mid-line nasal profile curve at its nasal root end
Prosthion	The lowest, most anterior point on the alveolar portion of the premaxilla (equivalent to the gum), in the mid-sagittal plane, between the upper central incisors
Infradentale	The anterior superior point on the mandible at its labial contact on the mid-sagittal plane
Promenton	The most projecting point of the chin in the mid-sagittal plane
Menton	The midpoint on the lower border of the mandible in the mid-sagittal plane
Frontotemporale (L, R)	The most anterior point on either side of the temporal crest of the minimum transverse breadth of the frontal bone
Supraorbitale (L, R)	The most superior point along the midline of the orbit on each superior orbital rim

Name	Definition
Ectoconchion (L, R)	The most lateral point on the orbital margin. Lies on the summit of the anterior convexity of the outer rim of the orbit, and not on its inner edge
Alare (L, R)	The most lateral point on each nasal's margin where maximum nasal breadth
Zygion (L, R)	The most lateral point on the outer edge of the outline of the zygomatic arch
Condylion Laterale (L, R)	The most lateral point on each mandibular condyle
Gonion (L, R)	The point on each lower human jaw closest to the vertex of the gonial angle
Opision	The most posterior mid-sagittal point on the posterior margin of the foramen magnum
Basion	The most anterior point of the inferior rim of the internal circumference of the great foramen magnum in the sagittal plane
Opisthocranium	The most posteriorly protruding point on the back of the skull, located in the Mid-sagittal plane
Zygofrontale (L, R)	The most lateral point of the frontal bone on its zygomatic process

A rigorous protocol was developed to ensure the reliability and validity of anatomically derived and spline-derived landmark placement. The Frankfurt plane was used as a reference to define the local coordinate system for each subject, which is then aligned to a consistent global coordinate system for the landmarking procedure. The Frankfurt plane is a plane passing through the highest point on the upper margin of the opening of each external auditory canal and the lowest point on the lower margin of the left orbit. The Porion and Infraorbitale landmarks were first identified to establish the Frankfurt plane. The mid-Porion point, a point bisecting the line connecting the left Porion and the right Porion points, was taken as reference for the origin of the Frankfurt plane, the left Porion was taken as reference for the positive Y-axis and the left Intraorbital was taken as reference for the X-Y plane. Once each skull was aligned to a standardized view based on the Frankfurt plane, the remaining 28 anatomical landmarks were manually identified.

To extract “Spline-derived landmarks” on the outer skull surface, a set of planes, including the mid-sagittal plane and 6 evenly distributed planes perpendicular to the mid-sagittal plane and passing through both left and right Porion points were identified. The intersection lines between the skull outer surface and the planes were used to define cubic splines for landmark extraction. Two additional splines connecting the Opision point to the most posterior coronal spline were used to capture the inferior-posterior region of the skull geometry. Evenly distributed landmarks were extracted on each cubic spline to identify a total of 72 spline-derived landmarks for each subject. Illustration of the intersection lines and spline-derived landmarks can be seen in [Fig.A1](#). Specifically, nine landmarks were evenly distributed on each of the six cross-sectional lines

between the outer skull and the six planes perpendicular to the mid-sagittal plane, and six landmarks were evenly distributed on each of the lines connecting the Opision point to the most posterior coronal spline.

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